



Controlling Switch Access with Passwords and Privilege Levels

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Restrictions for Controlling Switch Access with Passwords and Privileges

The following are the restrictions for controlling switch access with passwords and privileges:

- Disabling password recovery will not work if you have set the switch to boot up manually by using the **boot manual** global configuration command. This command produces the boot loader prompt (*switch:*) after the switch is power cycled.
- Password validation for the **enable password** command against the common criteria policy does not happen during configuration or reconfiguration of the **aaa common-criteria policy** command. The password is validated against the common criteria policy only during configuration or reconfiguration of the **enable common-criteria-policy** command.

In a high availability setup, if the active device is reloaded and then one of the criterion under the AAA common criteria policy associated with the enable password configuration is changed (such that the password no longer satisfies the common criteria policy) at a time instance between the manual reload of the active device and selection of the standby switch, the enable password configuration on the standby device fails during bulk sync, while the enable password configuration continues to exist on the active device. This configuration mismatch between the active and the standby devices triggers continuous reload of the standby device. We recommend that you do not modify the common criteria policy at a time instance between the manual reload of the active device and the standby switch selection.

Restrictions and Guidelines for Reversible Password Types

- Password type 0 and 7 are replaced with password type 6. So password type 0 and 7, which were used for administrator login to the console, Telnet, SSH, webUI, and NETCONF must be migrated to password type 6. No action is required if username and password are type 0 and 7 for local authentication such as CHAP, EAP, and so on.
- If the startup configuration has a type 6 password and you downgrade to a version in which type 6 password is not supported, you can/may be locked out of the device.

Restrictions and Guidelines for Irreversible Password Types

- Username secret password type 5 and enable secret password type 5 must be migrated to the stronger password type 8 or 9. For more information, see [Protecting Enable and Enable Secret Passwords with Encryption, on page 7](#).
- If the startup configuration of the device has convoluted type 9 secret (password that starts with \$14\$), then a downgrade can only be performed to a release in which the convoluted type 9 secret is supported. Convoluted type 9 secret is supported in Cisco IOS XE Gibraltar 16.11.2 and later releases. If the startup configuration has a convoluted type 9 secret, and you downgrade to a release prior to Cisco IOS XE Gibraltar 16.11.2, you can/may be locked out of the device.

Before you downgrade to any release in which convoluted type 9 secret is not supported, ensure that the type 9 secret (password that starts with \$9\$) must be part of the startup configuration instead of convoluted type 9 secret (password that starts with \$14\$) or type 5 secret (password that starts with \$1\$).

If a device is upgraded from Cisco IOS XE Fuji 16.9.x, Cisco IOS XE Gibraltar 16.10.x, or Cisco IOS XE Gibraltar 16.11.x to Cisco IOS XE Gibraltar 16.12.x, the type 5 secret is auto-converted to convoluted type 9 secret (password that starts with \$14\$). For example: `username user1 secret 5 $1$dNmW$7jWhqdtZ2qBVz2R4CSZZC0` is auto-converted to `username user1 secret 9 $14$dNmW$QykGZEEGmiEGrE$C9D/fD0czicOtgaZAa1CTa2sgygi0Leyw3/cLqPY426`. After the device is upgraded, run the **write memory** command in privileged EXEC mode for the convoluted type 9 secret to be permanently written into the startup configuration.

- Plain text passwords are converted to nonreversible encrypted password type 9.



Note This is supported in Cisco IOS XE Gibraltar 16.10.1 and later releases.

- Secret password type 4 is not supported.

Information About Controlling Switch Access with Passwords and Privileges

This section provides information about controlling switch access with passwords and privileges.

Preventing Unauthorized Access

You can prevent unauthorized users from reconfiguring your switch and viewing configuration information. Typically, you want network administrators to have access to your switch while you restrict access to users who dial from outside the network through an asynchronous port, connect from outside the network through a serial port, or connect through a terminal or workstation from within the local network.

To prevent unauthorized access into your switch, you should configure one or more of these security features:

- At a minimum, you should configure passwords and privileges at each switch port. These passwords are locally stored on the switch. When users attempt to access the switch through a port or line, they must enter the password specified for the port or line before they can access the switch.
- For an additional layer of security, you can also configure username and password pairs, which are locally stored on the switch. These pairs are assigned to lines or ports and authenticate each user before that user can access the switch. If you have defined privilege levels, you can also assign a specific privilege level (with associated rights and privileges) to each username and password pair.
- If you want to use username and password pairs, but you want to store them centrally on a server instead of locally, you can store them in a database on a security server. Multiple networking devices can then use the same database to obtain user authentication (and, if necessary, authorization) information.
- You can also enable the login enhancements feature, which logs both failed and unsuccessful login attempts. Login enhancements can also be configured to block future login attempts after a set number of unsuccessful attempts are made.

Default Password and Privilege Level Configuration

A simple way of providing terminal access control in your network is to use passwords and assign privilege levels. Password protection restricts access to a network or network device. Privilege levels define what commands users can enter after they have logged into a network device.

This table shows the default password and privilege level configuration.

Table 1: Default Password and Privilege Levels

Feature	Default Setting
Enable password and privilege level	No password is defined. The default is level 15 (privileged EXEC level). The password is not encrypted in the configuration file.
Enable secret password and privilege level	No password is defined. The default is level 15 (privileged EXEC level). The password is encrypted before it is written to the configuration file.
Line password	No password is defined.

Additional Password Security

The following sections provide information about unmasked and masked secret password.

Unmasked Secret Password

To provide an additional layer of security, particularly for passwords that cross the network or that are stored on a Trivial File Transfer Protocol (TFTP) server, you can use either the **enable password** or **enable secret** global configuration commands. Both commands accomplish the same thing; that is, you can establish an encrypted password that users must enter to access privileged EXEC mode (the default) or any privilege level you specify.

We recommend that you use the **enable secret** command because it uses an improved encryption algorithm. If you configure the **enable secret** command, it takes precedence over the **enable password** command; the two commands cannot be in effect simultaneously.

For a device that loads with no start-up configuration, the Enable Secret Password task is a mandatory configuration whether you select **Yes** or **No** at the "Would you like to enter the initial configuration dialog?" prompt of the initial configuration wizard. The configured password must contain a minimum of 10 and a maximum of 32 characters. It must also include a minimum of one uppercase letter, one lowercase letter, and one numeral. Additionally, the term 'cisco' must not be part of the password.



Note In some cases where the device is connected to the internet, Cisco Plug and Play (PnP) can terminate the initial configuration wizard. In such cases, the enable secret configuration will not be prompted.

If you enable password encryption, it applies to all passwords including username passwords, authentication key passwords, the privileged command password, and console and virtual terminal line passwords.

Masked Secret Password

With **enable secret** command, password is encrypted but is visible on the terminal when you type the password. To mask the password on the terminal, use the **masked-secret** global configuration command. The encryption type for this password is type 9, by default.

You can use this command to configure masked secret password for common criteria policy.

Password Recovery

By default, any end user with physical access to the switch can recover from a lost password by interrupting the boot process while the switch is powering on and then by entering a new password.

The password-recovery disable feature protects access to the switch password by disabling part of this functionality. When this feature is enabled, the end user can interrupt the boot process only by agreeing to set the system back to the default configuration. With password recovery disabled, you can still interrupt the boot process and change the password, but the configuration file (config.text) and the VLAN database file (vlan.dat) are deleted.

If you disable password recovery, we recommend that you keep a backup copy of the configuration file on a secure server in case the end user interrupts the boot process and sets the system back to default values. Do not keep a backup copy of the configuration file on the switch. If the switch is operating in VTP transparent mode, we recommend that you also keep a backup copy of the VLAN database file on a secure server. When the switch is returned to the default system configuration, you can download the saved files to the switch by using the Xmodem protocol.

To re-enable password recovery, use the **no system disable password recovery switch number | all** global configuration command.

Terminal Line Telnet Configuration

When you power-up your switch for the first time, an automatic setup program runs to assign IP information and to create a default configuration for continued use. The setup program also prompts you to configure your switch for Telnet access through a password. If you did not configure this password during the setup program, you can configure it when you set a Telnet password for a terminal line.

Username and Password Pairs

You can configure username and password pairs, which are locally stored on the switch. These pairs are assigned to lines or ports and authenticate each user before that user can access the switch. If you have defined privilege levels, you can also assign a specific privilege level (with associated rights and privileges) to each username and password pair.

Privilege Levels

Cisco devices use privilege levels to provide password security for different levels of switch operation. By default, the Cisco IOS XE software operates in two modes (privilege levels) of password security: user EXEC (Level 1) and privileged EXEC (Level 15). You can configure up to 16 hierarchical levels of commands for each mode. By configuring multiple passwords, you can allow different sets of users to have access to specified commands.

Privilege Levels on Lines

Users can override the privilege level you set using the **privilege level** line configuration command by logging in to the line and enabling a different privilege level. They can lower the privilege level by using the **disable** command. If users know the password to a higher privilege level, they can use that password to enable the higher privilege level. You might specify a high level or privilege level for your console line to restrict line usage.

For example, if you want many users to have access to the **clear line** command, you can assign it level 2 security and distribute the level 2 password fairly widely. But if you want more restricted access to the **configure** command, you can assign it level 3 security and distribute that password to a more restricted group of users.

Command Privilege Levels

When you set a command to a privilege level, all commands whose syntax is a subset of that command are also set to that level. For example, if you set the **show ip traffic** command to level 15, the **show** commands and **show ip** commands are automatically set to privilege level 15 unless you set them individually to different levels.

AES Password Encryption and Master Encryption Keys

You can enable strong, reversible 128-bit Advanced Encryption Standard (AES) password encryption, also known as type 6 encryption. To start using type 6 encryption, enable the AES Password Encryption feature and configure a master encryption key to encrypt and decrypt passwords.

After you enable AES password encryption and configure a master key, all the existing and newly created cleartext passwords for the supported applications are stored in type 6 encrypted format, unless you disable type 6 password encryption. You can also configure the device to convert all the existing weakly encrypted passwords to type 6 encrypted passwords.

Type 6 encrypted password that is configured must be compatible with the master key existing on the private NVRAM of the device. If it is not compatible, the configuration fails.

Type 0 and 7 passwords can be autoconverted to type 6 if the AES Password Encryption feature and master encryption key are configured.


Note

- Type 6 CLI support for TACACS and RADIUS related keys is supported from Cisco IOS XE Everest 16.5.x and later releases.
- Type 6 encrypted password for the username password is supported from Cisco IOS XE Gibraltar 16.10.1 and later releases. Autoconversion to password type 6 is supported from Cisco IOS XE Gibraltar 16.11.1 and later releases.
- Type 6 encrypted password for the username password is backward compatible to Cisco IOS XE Gibraltar 16.10.x. If you downgrade to any release earlier than Cisco IOS XE Gibraltar 16.10.1, the type 6 encrypted username password is rejected. After autoconversion, to prevent an administrator password from getting rejected during a downgrade, migrate the passwords used for administrator logins (management access) to irreversible password types manually.
- Type 6 encrypted password for enable password and line VTY password are supported from Cisco IOS XE Dublin 17.10.1 and later releases. Autoconversion to password type 6 is supported from Cisco IOS XE Dublin 17.11.1 and later releases.

How to Configure Switch Access with Passwords and Privileges

The following sections provide information about the various tasks to access the switch with passwords and privileges.

Setting or Changing a Static Enable Password

The enable password controls access to the privileged EXEC mode. Follow these steps to set or change a static enable password:

Procedure

	Command or Action	Purpose
Step 1	enable Example: Device> enable	Enables privileged EXEC mode. Enter your password, if prompted.
Step 2	configure terminal Example: Device# configure terminal	Enters global configuration mode.
Step 3	enable [common-criteria-policy policy-name] password password	Defines a new password or changes an existing password for access to privileged EXEC mode.

	Command or Action	Purpose
	Example: Device(config)# enable password secret321	- By default, no password is defined. - For <i>policy-name</i>, specify a policy name defined using the aaa common-criteria policy command. Note aaa new-model and aaa common-criteria policy commands must be configured before attaching the common-criteria-policy option to the password. - For <i>password</i>, specify a string from 1 to 25 alphanumeric characters. The string cannot start with a number, is case sensitive, and allows spaces, but ignores leading spaces. It can contain the question mark (?) character if you precede the question mark with the key combination Crtl-v when you create the password. For example, to create the password abc?123, do this: - Enter abc. - Enter Crtl-v. - Enter ?123. When the system prompts you to enter the enable password, you need not precede the question mark with Crtl-v; you can simply enter abc?123 at the password prompt.
Step 4	end Example: Device(config)# end	Exits global configuration mode and returns to privileged EXEC mode.

Protecting Enable and Enable Secret Passwords with Encryption

Follow these steps to establish an encrypted password that users must enter to access privileged EXEC mode (the default) or any privilege level you specify:

Procedure

	Command or Action	Purpose
Step 1	enable Example: Device> enable	Enables privileged EXEC mode. Enter your password, if prompted.
Step 2	configure terminal Example: Device# configure terminal	Enters global configuration mode.
Step 3	Use one of the following: • enable password [level level] {unencrypted-password encryption-type encrypted-password} • enable secret [level level] {unencrypted-password encryption-type encrypted-password} Example: <pre>Device(config)# enable password level 12 example123</pre> or <pre>Device(config)# enable secret 9 \$9\$sMLBsTFXLnnHTk\$0L82</pre>	• Defines a new password or changes an existing password for access to privileged EXEC mode. • Defines a secret password, which is saved using a nonreversible encryption method. • (Optional) For <i>level</i>, the range is from 0 to 15. Level 1 is normal user EXEC mode privileges. The default level is 15 (privileged EXEC mode privileges). • For <i>unencrypted-password</i>, specify a string from 1 to 25 alphanumeric characters. The string cannot start with a number, is case sensitive, and allows spaces but ignores leading spaces. By default, no password is defined. • For <i>encryption-type</i>, the available options for enable password are type 0, 6, and 7, and type 0, 5, 8, and 9 for enable secret. If you specify an encryption type, you must provide an encrypted password—an encrypted password that you copy from another switch configuration. Secret encryption type 9 is more secure, so we recommend that you select type 9 to avoid any issues while upgrading or downgrading. Note • If you do not specify an encryption type for the secret password, the password is auto converted to type 9. This is applicable in Cisco IOS XE

	Command or Action	Purpose
		Gibraltar 16.10.1 and later releases. - If you specify an encryption type and then enter a clear text password, it will result in an error. - You can also configure type 9 encryption for the secret password manually by using the algorithm-type script command in global configuration mode. For example: <pre>Device(config)# username user1 algorithm-type script secret cisco</pre> Or <pre>Device(config)# enable algorithm-type script secret cisco</pre> Run the write memory command in privileged EXEC mode for the type 9 secret to be permanently written into the startup configuration.
Step 4	service password-encryption Example: <pre>Device(config)# service password-encryption</pre>	(Optional) Encrypts the password when the password is defined or when the configuration is written. Encryption prevents the password from being readable in the configuration file.
Step 5	end Example: <pre>Device(config)# end</pre>	Exits global configuration mode and returns to privileged EXEC mode.

Disabling Password Recovery

Follow these steps to disable password recovery to protect the security of your switch:

Before you begin

If you disable password recovery, we recommend that you keep a backup copy of the configuration file on a secure server in case the end user interrupts the boot process and sets the system back to default values. Do

not keep a backup copy of the configuration file on the switch. If the switch is operating in VTP transparent mode, we recommend that you also keep a backup copy of the VLAN database file on a secure server. When the switch is returned to the default system configuration, you can download the saved files to the switch by using the Xmodem protocol.

Procedure

	Command or Action	Purpose
Step 1	enable Example: Device> enable	Enables privileged EXEC mode. Enter your password, if prompted.
Step 2	configure terminal Example: Device# configure terminal	Enters global configuration mode.
Step 3	system disable password recovery switch {all <1-9>} Example: Device(config)# system disable password recovery switch all	Disables password recovery. • <i>all</i>: Sets the configuration on switches in stack. • <i><1-9></i>: Sets the configuration on the switch number selected. This setting is saved in an area of the flash memory that is accessible by the boot loader and the Cisco IOS image, but is not a part of the file system and is not accessible by any user.
Step 4	end Example: Device(config)# end	Exits global configuration mode and returns to privileged EXEC mode.

What to do next

To remove **disable password recovery**, use the **no system disable password recovery switch all** global configuration command.

Setting a Telnet Password for a Terminal Line

Beginning in user EXEC mode, follow these steps to set a Telnet password for the connected terminal line:

Before you begin

- Attach a PC or workstation with emulation software to the switch console port, or attach a PC to the Ethernet management port.
- The default data characteristics of the console port are 9600, 8, 1, no parity. You might need to press the Return key several times to see the command-line prompt.

Procedure

	Command or Action	Purpose
Step 1	enable Example: Device> enable	Enables privileged EXEC mode. Enter your password, if prompted.
Step 2	configure terminal Example: Device# configure terminal	Enters global configuration mode.
Step 3	line vty 0 98 Example: Device(config)# line vty 0 98	Configures the number of Telnet sessions (lines), and enters line configuration mode. There are 99 possible sessions on a command-capable device. The 0 and 98 mean that you are configuring all 99 possible Telnet sessions.
Step 4	password { <i>unencrypted-password</i> <i>encryption-type encrypted-password</i> } Example: Device(config-line)# password 6 VeGSWf_NXfZOBPROFXVEaRf[TFeAAB	Sets a Telnet password for the line or lines. For <i>encryption-type</i> , enter 0 to specify that an unencrypted password will follow. Enter 7 to specify that a hidden password will follow. Enter 6 to specify that an encrypted password will follow.
Step 5	end Example: Device(config-line)# end	Returns to privileged EXEC mode.

Configuring Username and Password Pairs

Follow these steps to configure username and password pairs:

Procedure

	Command or Action	Purpose
Step 1	enable Example: Device> enable	Enables privileged EXEC mode. Enter your password, if prompted.
Step 2	configure terminal Example: Device# configure terminal	Enters global configuration mode.

	Command or Action	Purpose
Step 3	username <i>name</i> [privilege level] { password encryption-type password } Example: <pre>Device(config)# username adamsample privilege 1 password secret456 Device(config)# username 11111111111 mac attribute</pre>	Sets the username, privilege level, and password for each user. - For <i>name</i>, specify the user ID as one word or the MAC address. Spaces and quotation marks are not allowed. - You can configure a maximum of 12000 clients each, for both username and MAC filter. (Optional) For <i>level</i>, specify the privilege level the user has after gaining access. The range is 0 to 15. Level 15 gives privileged EXEC mode access. Level 1 gives user EXEC mode access. - For <i>encryption-type</i>, enter 0 to specify that an unencrypted password will follow. Enter 7 to specify that a hidden password will follow. Enter 6 to specify that an encrypted password will follow. - For <i>password</i>, specify the password the user must enter to gain access to the device. The password must be from 1 to 25 characters, can contain embedded spaces, and must be the last option specified in the username command.
Step 4	Use one of the following: line console 0 line vty 0 98 Example: <pre>Device(config)# line console 0</pre> or <pre>Device(config)# line vty 0 98</pre>	Enters line configuration mode, and configures the console port (line 0) or the VTY lines (line 0 to 98).
Step 5	end Example: <pre>Device(config-line)# end</pre>	Exits line configuration mode and returns to privileged EXEC mode.

Setting the Privilege Level for a Command

Follow these steps to set the privilege level for a command:

Procedure

	Command or Action	Purpose
Step 1	enable Example: Device> enable	Enables privileged EXEC mode. Enter your password, if prompted.
Step 2	configure terminal Example: Device# configure terminal	Enters global configuration mode.
Step 3	privilege mode level level command Example: Device(config)# privilege exec level 14 configure	Sets the privilege level for a command. - For <i>mode</i>, enter configure for global configuration mode, exec for EXEC mode, interface for interface configuration mode, or line for line configuration mode. - For <i>level</i>, the range is from 0 to 15. Level 1 is for normal user EXEC mode privileges. Level 15 is the level of access permitted by the enable password. - For <i>command</i>, specify the command to which you want to restrict access.
Step 4	enable password level level password Example: Device(config)# enable password level 14 SecretPswd14	Specifies the password to enable the privilege level. - For <i>level</i>, the range is from 0 to 15. Level 1 is for normal user EXEC mode privileges. - For <i>password</i>, specify a string from 1 to 25 alphanumeric characters. The string cannot start with a number, is case sensitive, and allows spaces but ignores leading spaces. By default, no password is defined.
Step 5	end Example: Device(config)# end	Exits global configuration mode and returns to privileged EXEC mode.

Changing the Default Privilege Level for Lines

Follow these steps to change the default privilege level for the specified line:

Procedure

	Command or Action	Purpose
Step 1	enable Example: Device> enable	Enables privileged EXEC mode. Enter your password if prompted.
Step 2	configure terminal Example: Device# configure terminal	Enters global configuration mode.
Step 3	line vty line Example: Device(config)# line vty 10	Selects the virtual terminal line on which to restrict access.
Step 4	privilege exec level level Example: Device(config-line)# privilege exec level 15	Changes the default privilege level for the line. For <i>level</i> , the range is from 0 to 15. Level 1 is for normal user EXEC mode privileges. Level 15 is the level of access permitted by the enable password.
Step 5	end Example: Device(config-line)# end	Exits line configuration mode and returns to privileged EXEC mode.

What to do next

Users can override the privilege level you set using the **privilege level** line configuration command by logging in to the line and enabling a different privilege level. They can lower the privilege level by using the **disable** command. If users know the password to a higher privilege level, they can use that password to enable the higher privilege level. You might specify a high level or privilege level for your console line to restrict line usage.

Logging in to and Exiting a Privilege Level

Beginning in user EXEC mode, follow these steps to log into a specified privilege level and exit a specified privilege level.

Procedure

	Command or Action	Purpose
Step 1	enable level Example: Device> enable 15	Logs in to a specified privilege level. In the example, Level 15 is privileged EXEC mode.

	Command or Action	Purpose
		For <i>level</i> , the range is 0 to 15.
Step 2	disable <i>level</i> Example: Device# disable 1	Exits to a specified privilege level. In the example, Level 1 is user EXEC mode. For <i>level</i> , the range is 0 to 15.

Configuring an Encrypted Preshared Key

To configure an encrypted preshared key, perform the following steps.

Procedure

	Command or Action	Purpose
Step 1	enable Example: Device> enable	Enables privileged EXEC mode. Enter your password if prompted.
Step 2	configure terminal Example: Device# configure terminal	Enters global configuration mode.
Step 3	key config-key password-encrypt [<i>text</i>] Example: Device(config)# key config-key password-encrypt	Stores a type 6 encryption key in private NVRAM. - To key in interactively (using the Enter key) and an encrypted key already exists, you will be prompted for the following: Old key, New key, and Confirm key. - To key in interactively, but an encryption key is not present, you will be prompted for the following: New key and Confirm key. - When removing the password that is already encrypted, you will see the following prompt: WARNING: All type 6 encrypted keys will become unusable. Continue with master key deletion? [yes/no]:
Step 4	password encryption aes Example: Device(config)# password encryption aes	Enables the encrypted preshared key.

	Command or Action	Purpose
Step 5	end Example: Device(config)# end	Exits global configuration mode and returns to privileged EXEC mode.

Monitoring Switch Access with Passwords and Privileges

Table 2: Commands for Displaying Privilege-Level Information

Command	Information
show privilege	Displays the privilege level configuration.

Configuration Examples for Switch Access with Passwords and Privilege Levels

Example: Setting or Changing a Static Enable Password

The following example shows how to change the enable password to *11u2c3k4y5*. The password is not encrypted and provides access to level 15 (traditional privileged EXEC mode access):

```
Device> enable
Device# configure terminal
Device(config)# enable password 11u2c3k4y5
Device(config)# end
```

Example: Protecting Enable and Enable Secret Passwords with Encryption

The following example shows how to configure the encrypted password *\$9\$sMLBsTFXLnnHTk\$0L82* for privilege level 2:

```
Device> enable
Device# configure terminal
Device(config)# enable secret level 2 9 $9$sMLBsTFXLnnHTk$0L82
Device(config)# end
```

Example: Setting a Telnet Password for a Terminal Line

The following example shows how to set the Telnet password to *let45me67in89*:

```
Device> enable
Device# configure terminal
Device(config)# line vty 10
Device(config-line)# password let45me67in89
Device(config-line)# end
```


Example: Setting the Privilege Level for a Command

The following example shows how to set the **configure** command to privilege level 14 and define *SecretPswd14* as the password users must enter to use level 14 commands:

```
Device> enable
Device# configure terminal
Device(config)# privilege exec level 14 configure
Device(config)# enable password level 14 SecretPswd14
Device(config)# end
```

Example: Configuring an Encrypted Preshared Key

The following example shows a configuration for which a type 6 preshared key has been encrypted. It includes the prompts and messages that a user might see.

```
Device> enable
Device# configure terminal
Device(config)# password encryption aes
Device(config)# key config-key password-encrypt
New key:
Confirm key:
Device(config)#
01:46:40: TYPE6_PASS: New Master key configured, encrypting the keys with
the new master key
Device(config)# end
```

Feature History for Controlling Switch Access with Passwords and Privileges

This table provides release and related information for the features explained in this module.

These features are available in all the releases subsequent to the one they were introduced in, unless noted otherwise.

Release	Feature	Feature Information
Cisco IOS XE Gibraltar 16.11.1	Controlling Switch Access with Passwords and Privileges	Password protection restricts access to a network or network device. Privilege levels define what commands users can enter after they have logged into a network device.
Cisco IOS XE Gibraltar 16.11.1	Autoconversion of Type 0 and Type 7 Username Password to Type 6	From this release, type 0 and 7 username password can be autoconverted to type 6.
Cisco IOS XE Cupertino 17.7.1	Enforce to Change the Administrator Default Password Function at first access to Device and to the Service Shell	For a device that loads with no start-up configuration, the Enable Secret Password task is a mandatory configuration in the initial configuration wizard.

Release	Feature	Feature Information
Cisco IOS XE Cupertino 17.7.1	Controlling Switch Access with Passwords and Privileges	This feature was implemented on the Cisco Catalyst 9600 Series Supervisor 2 Module (C9600X-SUP-2).
Cisco IOS XE Cupertino 17.8.1	Enforce Minimum Length for Enable Password	AAA common criteria policy support has been introduced in the enable password command.
Cisco IOS XE Dublin 17.11.1	Autoconversion of Type 0 and Type 7 Enable Password and Line VTY Password to Type 6	From this release, type 0 and 7 enable password and line vty password can be autoconverted to type 6.

Use the Cisco Feature Navigator to find information about platform and software image support. To access Cisco Feature Navigator, go to <https://cfng.cisco.com/>.